

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

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1. How many sequence of length 7 can you make using digits 1-8? 1 / 1 point

- ☐ $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2$
- ☐ 56
- ☒ 8^7
- ☐ $8 + 7 + 6 + 5 + 4 + 3 + 2$

👍 Nice work

This is correct. There are seven positions and for each there are eight digits to choose from. So, it is $8 \times 8 \times 8 \times 8 \times 8 \times 8 \times 8$.

2. How many sequence of length 6-8 can you make using even digits: 0, 2, 4, 6, 8? 1 / 1 point

- ☐ $5^6 \cdot 5^7 \cdot 5^8$
- ☐ $8! + 7! + 6!$
- ☐ $6^5 + 7^5 + 8^5$
- ☒ $5^6 + 5^7 + 5^8$

👍 Nice work

This is correct. You should take the union of the cases: length 6, 7, 8. For length 6, there are six positions and for each there are five digits to choose from, which is 5^6 . For length 7, it is 5^7 and for length 8 it is 5^8 .

3. How many sequence of length 5 can you make, where the first four letters are from uppercase English letters followed by one digit? 1 / 1 point

- ☐ $26 \times 25 \times 24 \times 23 \times 10$
- ☒ $26^4 \cdot 10$
- ☐ $36^4 \cdot 10$
- ☐ $4 \times 26 \times 10$

👍 Nice work

This is correct. There are five positions. For the first four, we choose among 26 letters for each, and for the last position we choose among 10 digits. So totally, we will have $26 \times 26 \times 26 \times 26 \times 10$ sequences.

4. How many passwords are there? 1 / 1 point

It has to be four characters drawn from uppercase letters or digits. Every password must contain at least one uppercase letter.

- ☐ 26.10.9.8
- ☐ $36 \cdot 10^3$
- ☒ $36^4 - 10^4$
- ☐ $26 \cdot 10^3$

👍 Nice work

This is correct. All possible sequences would be $36 \times 36 \times 36 \times 36$, because there are four positions and we can choose among 26+10 letters and digits. But we should exclude those sequences which have no uppercase letters. Those sequences without uppercase letters are $10 \times 10 \times 10 \times 10$ (only contain digits). So, the overall is $36^4 - 10^4$.

5. How many passwords are there? 1 / 1 point

It has to be four to six characters drawn from digits. Every password must contain at least one from {3,6,9}.

- ☐ $3 \cdot 7^3 + 3 \cdot 7^4 + 3 \cdot 7^5$
- ☒ $10^4 + 10^5 + 10^6 - 7^4 - 7^5 - 7^6$
- ☐ $3 \cdot 10^3 + 3 \cdot 10^4 + 3 \cdot 10^5$
- ☐ $3 \cdot 10 \cdot 9 \cdot 8 + 3 \cdot 10 \cdot 9 \cdot 8 \cdot 7 + 3 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6$

👍 Nice work

This is correct. All possible sequences would be $10^4 + 10^5 + 10^6$, because we should consider the union of four, five and six digit-long sequences. But we should exclude those sequences without {3,6,9}. Those sequences without {3,6,9} are $7^4 + 7^5 + 7^6$ (we should consider the union of four, five and six digit-long sequences). So, the overall is $10^4 + 10^5 + 10^6 - (7^4 + 7^5 + 7^6)$.

6. How many positive integers less than 150 are divisible by either 5 or 7? 1 / 1 point

- ☐ 52
- ☐ 50
- ☐ 45
- ☒ 46

👍 Nice work

This is correct. There are $149/5 = 29$ integers less than 150 which are divisible by 5. There are $149/7 = 21$ integers less than 150 which are divisible by 7. There are $149/35 = 4$ integers less than 150 which are divisible by both 5 and 7. So, the overall is $29 + 21 - 4 = 46$.

7. How many positive integers less than 100 are divisible by either 2, 3 or 5? 1 / 1 point

- ☐ 69
- ☒ 73
- ☐ 75
- ☐ 70

👍 Nice work

This is correct.

Using Inclusion-Exclusion Principle

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

$$49 + 33 + 19 - 16 - 9 - 6 + 3 = 73.$$